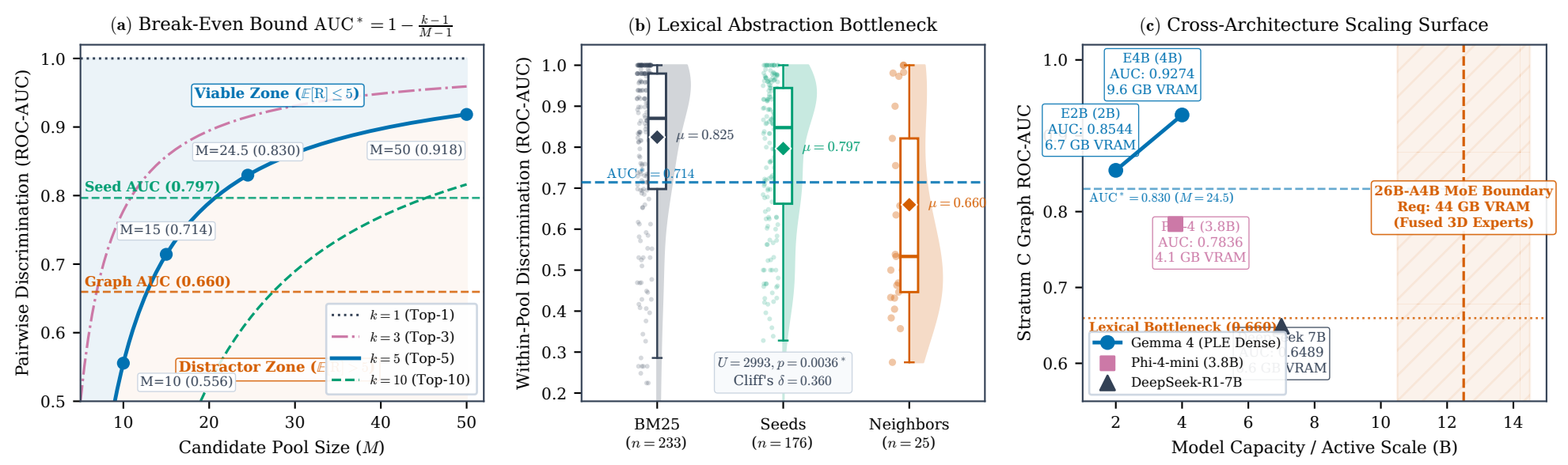

(a) Break-Even Bound AUC* = 1 − (k−1)/(M−1)
Viable Zone (E[R] ≤ 5)
M=24.5 (0.830)
M=50 (0.918)
Seed AUC (0.797)
M=15 (0.714)
Graph AUC (0.660)
Distractor Zone (E
M=10 (0.556)
k = 1 (Top-1)
k = 3 (Top-3)
k = 5 (Top-5)
k = 10 (Top-10)
Pairwise Discrimination (ROC-AUC)
Candidate Pool Size (M)
(b) Lexical Abstraction Bottleneck
μ = 0.825
μ = 0.797
μ = 0.660
AUC = 0.714
U = 2993, p = 0.0036 *
Cliff's δ = 0.360
BM25
(n = 233)
Seeds
(n = 176)
Neighbors
(n = 25)
Within-Pool Discrimination (ROC-AUC)
(c) Cross-Architecture Scaling Surface
E4B (4B)
AUC: 0.9274
9.6 GB VRAM
E2B (2B)
AUC: 0.8544
6.7 GB VRAM
AUC* = 0.830 (M = 24.5)
Phi-4 (3.8B)
AUC: 0.7836
4.1 GB VRAM
26B-A4B MoE Boundary
Req: 44 GB VRAM
(Fused 3D Experts)
Lexical Bottleneck (0.660)
Gemma 4 (PLE Dense)
Phi-4-mini (3.8B)
DeepSeek-R1-7B
Stratum C Graph ROC-AUC
Model Capacity / Active Scale (B)